

Document Assessment: What Remains Relevant in docs/

Hermes Agent (Lt. Cmdr Data), for Prof. Dr. J. Kapeller

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Version 4 (September 2026) **Version 3** (September 2026) **Version 2** (September 2026) **Version 1** (September 2026)

This document is the working plan for the Metroeconomica revision: Foundation I (labour-market closures), Foundation II (financial closures), the 5 x 3 evaluation matrix, and the staged workplan with integrated reviewer-response maps. It evolved from the document assessment of Versions 1–2, whose chronological review now lives in docs/archive/document_review_leftovers.md. Governing documents: ROADMAP.md and roadmaps/vertdict.md; active references: definitive_guide.md, labor_closures.md, and docs/reviews/.

Version 4 reports the intermediate implementation results of the notebook pipeline cbase2/ (v3 open-economy recalibration): the Stage 0 chapter is promoted to its own section with an updated status matrix, the remaining stages form the new Workplan section, and Stage 1 carries its updated state.

The document now opens with the two foundation sections whose decisions gate everything else – the labour-market closure (Foundation I) and the financing closure (Foundation II) – followed by the chronological workplan. All of this is part of the Version 2 reworking; the option tables in the foundation sections carry, per row, formulation, status, referee point, cost, and recommended role, plus the provenance tags (reviewer, old docs, Baqaee–Farhi usage).

1 Foundation I: Labour-Market Closure [added v2]

The referee demand (R2) is an explicit labour-supply elasticity; the code currently implements a geometric reallocation parameter and two wage regimes. The option tables below consolidate the full design space, grouped into the selected closures (BF plus the four standard alternatives ALPHA–DELTA), notable unselected candidates, and the dropped exogenous-shift closure. Provenance tags: "reviewer" attributes R1/R2; "old docs" names documents (archived under docs/archive/, plus the governing ROADMAP.md and roadmaps/vertdict.md); "used by BF" points to the Baqaee–Farhi papers (both source-verified in bf_replication/ and bf_replication2/):

- BF 2019 (Econometrica): Dokumente/Baqaee-2019-The Macroeconomic Impact of Microeconomic Shocks–Beyond Hulten's Theorem–Econometrica.pdf
- BF 2022 (AER): Dokumente/Baqaee-2022-Supply and Demand in Disaggregated Keynesian Economies with an Application to the COVID-19 Crisis–American Economic Review_4.pdf

1.1 Selected closures [added v2]

Suggested narrative: the paper measures how far the estimated effects of a sector-specific green-investment programme travel between the CGE and IO endpoints as the labour-market closure varies. Closure BF operationalizes the neoclassical short-run argument as a market friction: BF 2019's own framework contrasts full reallocation with none, and the η parameter makes that friction continuous – limits on intersectoral reallocation, not a change of paradigm, separate the short run from the benchmark. On the shared production-network core, four standard closures span the space: ALPHA fixes aggregate labour at full employment (the neoclassical standard); BETA relaxes the endowment through a labour–leisure elasticity (the voluntary response the referees ask for); GAMMA fixes the real wage and lets employment absorb the shock (production-agnostic Keynesian: the mechanism operates on the labour market rather than through the production function, so the extensive-margin result survives any production structure); DELTA fixes the real factor price with unconstrained quantity and fixed coefficients, emulating the Leontief inverse (Robinson 2006). The measurement claim: the wage regime (ALPHA/BETA vs. GAMMA) and the substitution elasticities govern the distance to DELTA; the reallocation friction BF is material but secondary; and the financing closure keeps every experiment admissible.

Closure BF: Geometric reallocation parameter (η, market friction)	<i>Endogenous sectoral allocation between immobile and cost-minimizing; the BF-comparable short-run friction</i>
Formulation	$L_i = L_{i,\text{fixed}}^{1-\eta} L_{i,\text{costmin}}^\eta$ with $\eta \in [0, 1]$ (extrapolation outside); total employment fixed at $\bar{L}$; not a labour-supply elasticity
Status	Implemented and tested (MobileLaborCES, :mobile); Sobol first-order share about 15.7%
Referee point addressed	The mobility margin only; total supply stays fixed
Cost	None
Recommended role	Selected: the BF-comparable market-friction closure; the interpolation parameter is reported under an honest name (“reallocation parameter”)
Mentioned by reviewer	both (R1.6 mobility/wage-rule question; R2.5 immobile-labour criticism – indirectly)
Mentioned in old docs	labor_closures.md; definitive_guide.md; WORKPLAN3.md (2026-09-03 addendum)
Used by BF	Yes at its endpoints – the original BF 2019 replication ships both variants: immobile labour (“no reallocation”; sector-specific wage vector, Simulation_Derivs.m) and mobile labour (“full reallocation”; single wage pinned at the numeraire, Simulation_Derivs_realloc.m; Read_Me.txt: “the first replicates results with immobile labor and the second replicates results with mobile labor”). The geometric interpolation parameter <i>between</i> the endpoints is project-specific
Closure ALPHA: Full-employment mobile labour (neoclassical standard)	<i>The standard full-employment CGE endpoint</i>
Formulation	$\sum_i L_i = \bar{L}$; one economy-wide wage; cost-minimizing allocation; the $\eta = 1$ limit of the reallocation parameter (as :mobile)
Status	Implemented (:mobile; ROADMAP Closure A); baseline endpoint of the wage-regime sweep
Referee point addressed	R2.5 (economy-wide mobile labour); R1.6 (unified wage rule enabling mobility)
Cost	None
Recommended role	Baseline endpoint of the closure continuum
Mentioned by reviewer	both (R2.5 explicitly recommends it; R1.6 asks for it)
Mentioned in old docs	REVISED_SALVAGE_PLAN.md; WORKPLAN.md; WORKPLAN2.md; verdictict.md; definitive_guide.md
Used by BF	Yes – BF 2019 baseline: competitive core with inelastically supplied (fixed) composite-factor endowment and uniform factor prices (see filename above)
Closure BETA: Elastic total labour supply (neoclassical, labour–leisure)	<i>Voluntary labour-supply response along the real wage; the continuum from vertical to horizontal supply</i>
Formulation	$L^s = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$ with a labour–leisure interpretation; $\eta_s = 0$ fixed supply; $\eta_s \rightarrow \infty$ approaches the fixed-real-wage limit
Status	Not implemented; specified in ROADMAP sections 4.2 and 5 (Closure B)
Referee point addressed	R2’s core demand (“the one obvious set of elasticities that really matters”)

Closure BETA: Elastic total labour supply (neoclassical, labour-leisure)	<i>Voluntary labour-supply response along the real wage; the continuum from vertical to horizontal supply</i>
Cost	Moderate: extends the $2N+1$ system by the supply equation and its anchor
Recommended role	Selected as the referee-facing principal addition
Mentioned by reviewer	both (R2 explicitly and centrally; R1 implicitly via the slack-scenario discussion)
Mentioned in old docs	REVISED_SALVAGE_PLAN.md (original elastic_labor_slack proposal); WORKPLAN.md; WORKPLAN2.md; PRELIMINARY-ASSESSMENT.md (baseline-anchoring correction); verdict.md (ElasticLaborSupply); ROADMAP.md
Used by BF	No (2019: inelastic endowments; 2022: sticky wages – verdict names the 2019 base as the right foundation for adding it)
Closure GAMMA: Sticky real wage, uncapped (production-agnostic Keynesian)	<i>Fixed real wage; employment absorbs the shock – the extensive margin</i>
Formulation	$w/P = \bar{w}$ (wage pinned at baseline); employment endogenous and uncapped; the gap $\bar{L} - \sum_i L_i$ is computed post-solve, not an equilibrium constraint
Status	Implemented; large positive GDP responses (+19.3 pp at mult 10, employment to 1.19x), residuals at most 1.9×10^{-7}
Referee point addressed	R1.5 (wage-setting rule: fixed real/nominal wage); R2's "fix the wage rate at baseline and drop the labour supply constraint" recipe
Cost	None
Recommended role	Report as-is, with the uncapped-employment caveat stated
Mentioned by reviewer	both (R2 gives the recipe explicitly; R1.5 demands the wage-setting rule)
Mentioned in old docs	WORKPLAN3.md (Milestone F + 2026-09-03 addendum); definitive_guide.md (Result 3); labor_closures.md
Used by BF	Yes (adapted) – BF 2022's sticky-wage mechanism (downward wage rigidity, demand-determined employment) in simplified one-wage form (see filename above)
Closure DELTA: IO endpoint (emulating the Leontief inverse)	<i>Fixed real factor price, unconstrained factor quantity; the exact conditions under which the CGE reproduces the IO multiplier</i>
Formulation	Fixed real primary-factor price; unconstrained factor quantity within the experiment; Leontief input coefficients (or a demonstrated CES special case with unchanged relative prices)
Status	Not implemented as an explicit closure (ROADMAP Closure D); GAMMA approximates its spirit
Referee point addressed	R2 explicitly (Robinson 2006: fix primary input prices, unconstrained supplies – the multiplier model); R1 explicitly (McGregor-Swales-Yin long-run zero-price-change IO-type result)
Cost	Small-moderate

Closure DELTA: IO endpoint (emulating the Leontief inverse)	<i>Fixed real factor price, unconstrained factor quantity; the exact conditions under which the CGE reproduces the IO multiplier</i>
Recommended role	Required: R2's entire critique turns on the exact conditions of equivalence
Mentioned by reviewer	both (R2 with the Robinson 2006 quote; R1 with McGregor et al. 1996)
Mentioned in old docs	verdict.md (FixedRealWage endpoint); ROADMAP.md (Closure D); definitive_guide.md (Part IV); REVISED_SALVAGE_PLAN.md (the superseded $\eta \rightarrow \infty$ endpoint idea)
Used by BF	No (origin is the IO tradition and Robinson 2006, not the BF papers; BF 2019 computes efficient reallocations, not fixed-price multipliers)

1.2 Notable candidates, but unselected [added v2]

Structural alternatives that did not make the selection: the partial-mobility 3N system (allocation-axis alternative) and the unemployment complementarity (optional, policy-evaluation only).

Standard partial-mobility formulation (3N system)	<i>Structural alternative: sectoral wages with wage-responsive sectoral supply</i>
Formulation	$L_i^s = \bar{L} s_i^0 (w_i/\bar{w}_i)^\nu / \sum_j s_j^0 (w_j/\bar{w}_j)^\nu$; sectoral labour demand from Shephard's lemma; N zero-profit + $(N-1)$ market-clearing + N sectoral labour-market + one numeraire = $3N$ equations
Status	Open design proposal in labor_closures.md; not implemented
Referee point addressed	None directly; referee-adjacent (R1.6, R2.5)
Cost	Largest: a new equilibrium system
Recommended role	Fallback if the one-wage supply formulation of Closure B is judged too reduced-form
Mentioned by reviewer	no
Mentioned in old docs	labor_closures.md (the proposal, with references: Hsieh-Klenow 2009; Baqaee-Farhi 2020 QJE; Artuc-Chaudhuri-McLaren 2010)
Used by BF	No (closest analogue: BF 2022's N sector-specific labour markets, but those are sticky, not supply-responsive)

Closure C: Unemployment complementarity (capped)	<i>Employment capped at $\bar{L}$; full-employment or wage-floor regime switches endogenously</i>
Formulation	$0 \leq \bar{L} - L \perp w/P - \bar{\omega} \geq 0$: complementarity between the employment gap and the real-wage gap
Status	Not implemented (ROADMAP section 5, Closure C; definitive guide Part IV records "partially")
Referee point addressed	R1.5 (wage-setting rule, wage curve, involuntary unemployment); R2 (DDR/Robinson unemployment closures; skill-class split)
Cost	Larger: complementarity solver, new results
Recommended role	Optional; mandatory only if the paper claims policy evaluation (ROADMAP Phase 6)
Mentioned by reviewer	both (R1.5 wage curve; R2's closure-literature demands)
Mentioned in old docs	verdict.md (UnemploymentComplementarity); ROADMAP.md; definitive_guide.md (Part IV status)

Closure C: Unemployment complementarity (capped)	<i>Employment capped at $\bar{L}$; full-employment or wage-floor regime switches endogenously</i>
Used by BF	Yes (adapted) – BF 2022’s complementarity machinery for binding wage floors / labour capacity, to be stripped down; its N -sector sticky-labour structure must not be copied unchanged (verdict; see filename above)

1.3 Dropped [added v2]

Regretfully retired: the exogenous labour-endowment shift in its legacy-callback implementation – the original paper’s own mechanism. The lineage note below records its direct resonance with BF 2022 and with the response letter’s methodological story.

Dropped: Exogenous labour-endowment shift (the legacy labour-slack callback)	<i>The original paper’s own mechanism: exogenous sectoral slack vectors – the same closure BF 2022 calibrates as labour-supply shifts</i>
Formulation	Sectoral labour endowments shifted exogenously: $\bar{L}_i \rightarrow \bar{L}_i(1 + g_i)$ with calibrated or assumed g_i ; no wage response required. Code implementations: full_labor_slack / full_labor_slack_alt (Leontief-projection sectoral vectors, manuscript Section 5.1) and empirical_labor_slack (uniform $I_{\text{new}} = (1 - \mu)^{-1} I$, Section 5.2)
Status	Implemented (legacy); the original submission’s Sections 5.1–5.2; ROADMAP section 4.4 lists it as a distinct closure concept
Referee point addressed	R1.5 describes precisely this mechanism (“is it simply a matter of increasing labour supply by the number of unemployed persons?”); R2 criticizes its use as the central bridge (deus ex machina; R2.10 uniform allocation)
Cost	None
Recommended role	Dropped from the model; its economics survive in the lineage note below and in the response letter’s methodological story
Mentioned by reviewer	both (R1 describes the mechanism; R2 criticizes its use as deus ex machina)
Mentioned in old docs	REVISED_SALVAGE_PLAN.md; WORKPLAN.md; WORKPLAN2.md; labor_closures.md; definitive_guide.md; WORKPLAN3.md
Used by BF	Yes – BF 2022 calibrates exogenous sectoral labour-supply shifts to BLS May-2020 sectoral hours (contractions; bf_replication2/src/model.jl applies them on the labour block); run in reverse, the same closure is the slack injection. BF 2019 shocks factor endowments in its general framework

Lineage note (original manuscript). The resonance is direct, not analogical. The original submission’s Section 5 already framed labour slack as exactly this closure: “the introduction of such a labor slack by itself is to be conceived and interpreted as the advent of an exogenous shock ... a spontaneous expansion of the labor force.” Its legitimizing narrative was borrowed from BF 2022’s channel: “the context of Corona-related lockdowns that partially required people to abstain from work due to social distancing regulations” (with BF 2022’s minimum-wage unemployment explicitly set aside because it “does not relax capacity constraints”). The code implements the manuscript’s two variants: full_labor_slack and full_labor_slack_alt (Leontief-projection-calibrated sectoral vectors; Section 5.1, aggregate employment +1.7%) and empirical_labor_slack (the uniform rule $I_{\text{new}} = (1 - \mu)^{-1} I$ with μ the ILO unemployment rate of 3.2%; Section 5.2, +3.1%). The referees’ treatments map onto these two variants: R1 questions 5.1’s rationale (“unsure about the rationale behind this scenario”) and 5.2’s missing wage-setting rule; R2’s “deus ex machina” targets the injection as the bridge

mechanism and the uniform transfer as arbitrary (R2.10). BF 2022 runs the same machinery as calibrated contractions; the original manuscript ran it as calibrated expansions. The revision therefore changes not the closure’s existence but its role: from the central bridge mechanism to a named scenario closure, with the bridge carried by endogenous labour supply (Closure B), the wage regime, and an explicit financing closure.

1.4 Coverage check: BF closures outside our lists [added v2]

Verified against the two papers ([Dokumente/](#)) and the replications; it spans both foundation lists. BF 2019 itself uses nothing beyond Closure A: its factor supply is inelastic at the endowment (“the supply of the factor is inelastic”), factors are non-reproducible goods, and the paper explicitly makes “no allowance for amplification of shocks through endogenous labor supply and capital accumulation” – elastic labour supply (Closure B) is named there as a channel they do not use. BF 2022, however, uses four closure elements that our two foundation lists do not cover:

BF 2022 element	Coverage status in our lists
Downward nominal wage rigidity, sector-specific (N sticky labour factors)	Partially covered by Closure C (adapted, stripped down). Two differences: nominal floor vs. our real-wage floor (moot only because the static model has $CPI = 1$; it bites under deflation), and N sector-specific floors vs. one economy-wide floor. The replication implements sticky sectoral labour but not the nominal/deflation side
Credit constraints / hand-to-mouth vs. Ricardian households	Not covered. Demand-side income-distribution closure (who spends the shock); our model has one representative household; ROADMAP leaves it out of scope
Zero lower bound / monetary policy	Not covered. Nominal closure; absent from the local replication (which keeps only nominal GDP accounting) and outside our static real model; relevant only if the paper’s question shifts, per the verdict’s 2019-vs-2022 criterion
Multiple primary factors (capital alongside labour; flexible capital vs. sticky labour in BF 2022)	Not covered by a foundation table. Our model and the 2019 replication are one-factor (value added as composite); ROADMAP section 5 carries it as the “optional extension” (mandatory only for policy-evaluation claims)
Firm capacity constraints (social distancing)	Not covered. BF 2022 models a supply shock as a labour-supply shift or a firm capacity constraint (“firms could be forced to operate at lower capacity ... due to social distancing”) – a rationing element outside our price-based static system
Reallocation frictions (adjustment costs, kappa)	Not covered by a foundation table. BF 2019 Online Appendix 2 (“Stuck Intermediates and Adjustment Costs”, the kappa variants); ours is a static model – <code>labor_closures.md</code> names convex employment-adjustment costs for a dynamic extension

None of these gaps blocks the current plan: the first is exactly the “stripped-down 2022 closure” the verdict prescribes, and the remaining three enter only via the optional extensions already flagged in Stage 0 (scope decisions) and ROADMAP section 5. They are listed here so the response letter can state precisely which BF elements are and are not used. One replication-code nuance to verify before citing the BF 2019 mobile variant as Closure A exactly: in the MATLAB code that variant pins $w = 1$ (numeraire) and solves only the N price equations, so the aggregate labour constraint is implicit rather than imposed; the paper itself states inelastic aggregate factor supplies. The exogenous labour-endowment shift, though present in both BF papers, is a different closure from the sticky wage: it moves supply without a wage response, whereas `:fixed` moves the wage floor and lets employment respond.

2 Foundation II: Financial Closures [added v2]

The four admissible experiment types of ROADMAP Phase 2. The current `demand_shock` is an unfinanced add-on and inadmissible by the ROADMAP’s own rule until one principal and one robustness variant are chosen; the explicit demand anchor introduced with the financing closure is also expected to resolve the $\eta = 1$ stall (scale indeterminacy) observed

from multiplier 0.5 upward. Neither BF paper supplies a fiscal financing closure (verdict); only the compositional-shifter normalization has a BF 2022 analogue.

2.1 Selected [added v2]

The three financing variants carried into the evaluation matrix: the preference-reallocation/tax-financed bracket as principal, external debt as the third variant.

Budget-neutral preference reallocation	<i>Compositional household demand shift with the budget closed by construction</i>
Budget identity	$\tilde{\beta}_i = \beta_i d_i / \sum_j \beta_j d_j$; $Z(p) = \sum_j \tilde{\beta}_j p_j^{1-\sigma}$; $c_i^h = E^h \tilde{\beta}_i p_i^{-\sigma} / Z(p)$, so $\sum_i p_i c_i^h = E^h$ holds exactly
Economic interpretation	Compositional demand shift; the deduction runs through the household's consumption composition (other categories shrink within the fixed E^h); government spending untouched; must not be described as an autonomous investment multiplier (verdict warning)
Cost / status	Low: normalization is half-done already (Milestone A normalized household demand)
Recommended role	Robustness / diagnostic benchmark
Mentioned by reviewer	no (neither referee demands it; R1's demand-shock rationale question is adjacent)
Mentioned in old docs	verdict.md (redesign section 1); ROADMAP.md (Phase 2); definitive_guide.md
Used by BF	Yes (analogous treatment) – BF 2022 renormalizes compositional demand shifters so the budget holds exactly (bf_replication2/src/model.jl:442); but as a demand shifter, not a financing device (see filename above)
Tax-financed public investment	<i>Autonomous green investment with an explicit tax counterparty</i>
Budget identity	$\sum_i p_i g_i = T$ with a lump-sum or specified tax rule; separate government/investment vector g_i
Economic interpretation	Autonomous green investment financed by taxation
Cost / status	Medium: new g_i vector plus budget equation; not started
Recommended role	Principal (recommendation)
Mentioned by reviewer	no (the financing gap was identified internally; the referees demand shock transparency only)
Mentioned in old docs	verdict.md; ROADMAP.md (Phase 2); PRELIMINARY-ASSESSMENT.md (financing admissibility); definitive_guide.md (Part V)
Used by BF	No (verdict: neither model supplies the fiscal financing closure)
Debt/foreign-financed net expenditure	<i>Demand expansion with an intertemporal or external counterpart</i>
Budget identity	$\sum_i p_i g_i = B + F$ with a defined saving–investment or current-account rule

Debt/foreign-financed net expenditure	<i>Demand expansion with an intertemporal or external counterpart</i>
Economic interpretation	Net new expenditure against borrowing or the external balance. Static-horizon nuance: debt finance (B) is resource-equivalent to the lump-sum tax (the household's current consumption gives up the same resources; the distinction becomes real only in a dynamic setting), while foreign finance (F) is genuinely distinct – resources arrive via the external balance (imports serve the investment)
Cost / status	Medium–high: additional institutional balance
Recommended role	Third variant (external debt); B collapses into the tax variant at this horizon
Mentioned by reviewer	no
Mentioned in old docs	vertdict.md; ROADMAP.md (Phase 2); definitive_guide.md
Used by BF	No

Financing design (user decision): taking the housing stimulus as vantage point and ruling out unfinanced manna, the realistic financing lies between the preference-reallocation and tax-financed variants, so the principal experiments are a bracket of the two. A third variant with external debt (foreign financing, F) is added: within this static real model it is the closest representation of the endogenous-money intuition that a borrowed euro need not be withheld from domestic consumption ex ante – the purchasing power enters, and the ex-post identity $I = S$ (here: the external deficit) records where the resources came from. The model itself cannot represent endogenous money: with the nominal side collapsed it displays only the ex-post accounting, so the F variant is an accounting open-economy closure, not a monetary mechanism. Imports feedback: at the current pipeline the imported content of the g_i vector is fixed by sector import shares, so the external variant absorbs the demand through net imports mechanically; a behavioral feedback (relative prices shifting import shares) would require an Armington-type extension, for which the existing $\Omega_{\text{dom}}/\Omega_{\text{raw}}$ split is the natural hook. Deduction routes recap: preference reallocation deducts from the household's consumption composition, expenditure switching from old government expenditure, tax financing from household income. Recommendation: run the preference-reallocation/tax-financed bracket as principal, foreign financing as the third variant; debt (B) collapses into the tax variant at this horizon; expenditure switching remains available as a secondary option. The full 5 x 3 evaluation matrix is assembled in the next section.

Financing design (user decision): taking the housing stimulus as vantage point and ruling out unfinanced manna, the realistic financing lies between the preference-reallocation and tax-financed variants, so the principal experiments are a bracket of the two; external debt is added as the third selected variant. Deduction routes recap: preference reallocation deducts from the household's consumption composition, expenditure switching from old government expenditure, tax financing from household income via the lump-sum tax. Imports feedback: at the current pipeline the imported content of the g_i vector is fixed by sector import shares, so the external variant absorbs the demand through net imports mechanically; a behavioral feedback (relative prices shifting import shares) would require an Armington-type extension, for which the existing $\Omega_{\text{dom}}/\Omega_{\text{raw}}$ split is the natural hook. The full 5 x 3 evaluation matrix is assembled in the next section.

2.2 **Dropped [added v2]**

Regretfully retired as a separate variant: expenditure switching. Pure debt financing (B) is not given its own variant at all – within the static horizon it is resource-equivalent to the tax variant (the distinction becomes real only in a dynamic setting), so it would only re-label the principal tax experiment.

Expenditure-switching investment	<i>Reallocation within a fixed public budget</i>
Budget identity	$\sum_i p_i g_i = T$ with T raised by cutting other public spending
Economic interpretation	Public budget reallocated across sectors; the deduction runs through old government expenditure ; household budgets untouched; no new purchasing power
Cost / status	Low–medium
Recommended role	Robustness alternative
Mentioned by reviewer	no

Expenditure-switching investment	Reallocation within a fixed public budget
Mentioned in old docs	verdict.md; ROADMAP.md (Phase 2); definitive_guide.md
Used by BF	No

The endogenous-money issue. The motivation for the external-debt variant is the endogenous-money intuition: in a world where credit money is created alongside the loan, a borrowed euro need not be withheld from domestic consumption ex ante; the ex-post identity $I = S$ holds by accounting, but it does not describe an ex-ante fund that had to be taken from anyone. The static real model cannot represent this mechanism itself: with the nominal side collapsed there is no credit creation, no banking sector, no interest-rate channel – the model displays only the ex-post accounting. What the F variant does capture is the shadow of the intuition: purchasing power enters without an ex-ante sacrifice of domestic consumption, and the external deficit records where the resources came from. It should therefore be sold as an accounting open-economy closure, not as a monetary mechanism. The contrast between the tax-financed and external-debt variants is nonetheless paradigmatically productive: it operationalizes, inside one framework, the difference between a loanable-funds reading (domestic saving must finance investment ex ante) and an endogenous-money reading (finance precedes saving; the balance closes ex post) – exactly the kind of commensurability the revised paper set out to measure.

3 The Evaluation Matrix (5 x 3) [added v2]

Decision (user): three financial closures – the preference-reallocation and tax-financed bracket plus the external-debt variant – crossed with the five selected labour closures gives a 5 x 3 matrix of 15 model variants to evaluate. This is not overblown: each cell is one solve of a 71-sector static system, computationally trivial; the interpretive burden is managed by reporting one headline set per cell (Torqvist real GDP response, employment, external balance) with sectoral detail in an appendix. Two design amendments: BF's row needs a pre-registered η^* (proposal: 0.5, fixed before Stage 2 results are inspected), and DELTA is a corner of the matrix rather than an independent equilibrium row – its cells are reached as GAMMA plus Leontief technology. The matrix serves the paradigmatic main point directly: it turns the search for commensurability into a measured map, with financing closures setting the scale of the stimulus and labour closures its transmission margin.

Expected signatures, pre-registered before any Stage 2 run (headline quantities per cell; sectoral detail deferred):

Labour closure	Financing: preference reallocation	Financing: tax-financed	Financing: external debt F
BF (friction, η^*)	Composition shift under friction: aggregate ~ flat; sectoral mix most distorted	Tax withdrawal with frictional reallocation; GDP response smaller than ALPHA	Externally financed demand, frictional allocation; import leakage plus misallocation
ALPHA (full employment)	Composition only; aggregate ~ 0 (the η -invariance diagnostic); sectoral reallocation	Crowding-out via wages/prices; aggregate ~ 0 (classic full-employment CGE)	Demand leaks to imports; wage rise moderated; external deficit = F
BETA (elastic supply)	Composition only; employment ~ flat along the supply elasticity	Positive but modest; employment and wage share the adjustment (η_s -dependent)	Positive; import leakage dampens the domestic employment response
GAMMA (sticky real wage)	Composition shift; small positive employment response	The extensive-margin result: employment absorbs the shock (current: +19.3 pp at mult 10)	Employment absorbs the domestically-produced share; imports take the rest
DELTA (IO endpoint, corner)	IO multiplier on the compositional shift (value-added effects)	IO multiplier net of the tax withdrawal	Full IO multiplier with import-adjusted inverse; external deficit = F

Limit relations across the matrix (reported as results, not hidden as redundancy): BF at $\eta = 1$ coincides with ALPHA, and BF at $\eta = 0$ is the immobile benchmark; BETA's high η_s limit approaches GAMMA's fixed-real-wage regime; and

GAMMA plus Leontief technology is DELTA. DELTA is therefore a corner, not an independent row; BF's endpoints duplicate ALPHA and the immobile benchmark. The coincidences are the bridge result: the same endpoint is reached from different directions, which is precisely the paradigmatic-geography claim of the revised paper.

Paradigmatic directions towards the short run

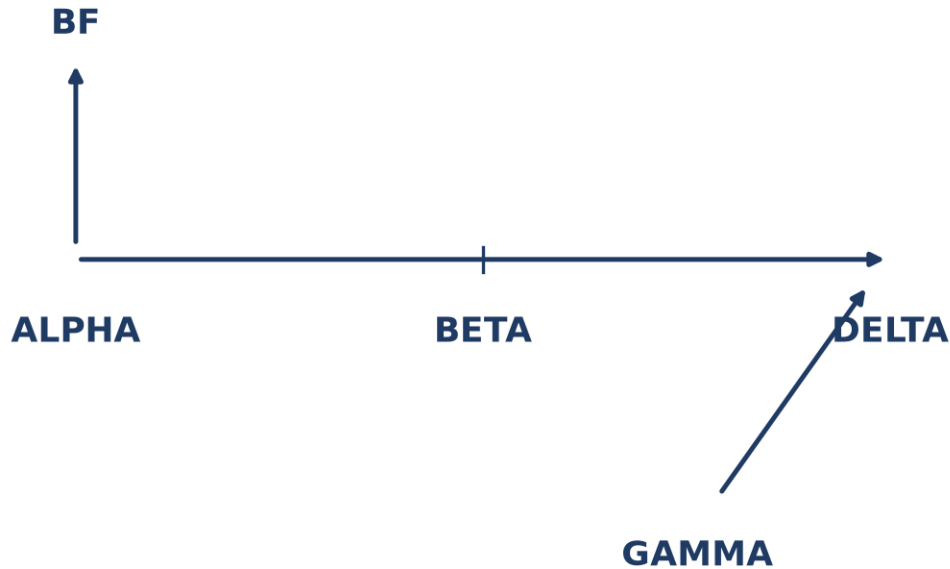


Figure 1: Figure: Paradigmatic directions towards the short run. The horizontal axis is the labour-supply-elasticity continuum (ALPHA at zero, BETA intermediate, DELTA at the IO limit); BF is the orthogonal reallocation-friction dimension anchored at the fixed-supply end; GAMMA approaches the IO endpoint from the wage-rigidity direction (GAMMA plus Leontief technology). Drafted after the user's sketch.

4 Stage 0: Conceptual decisions [reworked v3] [section 4 since v4]

Resolved in this round: the labour-closure selection (Foundation I: BF, ALPHA, BETA, GAMMA, DELTA), the financing design (Foundation II: the preference-reallocation/tax-financed bracket as principal, external debt as the third selected variant; expenditure switching dropped, pure debt subsumed by the tax variant), and the consolidation of the exogenous-shift/legacy closure as dropped. The status matrix below completes the Stage 0 answer by locating every cell of the evaluation matrix in the current implementation. Two items remain open: the audit-trail disclosure decision (Stage 4) and the scope locks (skill classes, open-economy depth, Stream C off the critical path).

The two paradigmatic wedges. The matrix has two axes, and both are paradigmatic wedges. The first – the labour-market closure – is the referees' axis: it sets the transmission margin of a demand stimulus, from allocation friction (BF) through full employment (ALPHA) and voluntary supply (BETA) to wage rigidity (GAMMA) and the IO endpoint (DELTA). The second – the financing closure – runs through the financing assumptions: it sets where the purchasing power of the stimulus comes from and through which budget it closes (household composition vs. household income vs. external balance), i.e. the loanable-funds versus endogenous-money contrast. Recommended strength of the second dimension: first-class as structure, bounded as claim. It is a full matrix axis and appears in every results table; but in the static real model it operates as accounting closures, so the paper claims the financing contrast as measured sensitivity, not as a theory of money.

Status matrix (Stage 0 answer). Symbols: ✔ implemented and validated; ● partially available – the machinery exists but needs admissibility re-runs, explicit framing, or hardening (the missing piece is stated); ● not implemented; 🔥 exists in the code but is inadmissible under the ROADMAP gates (the unfinanced autonomous shock). No cell is ✔ yet – under the corrected admissibility standard that is precisely the finding, and the matrix defines the Stage 1 work.

Labour closure	F1: preference reallocation	F2: tax-financed	F3: external debt
BF (friction, η^*)	● geometric η implemented; missing η^* pre-registration and the explicit $\tilde{\beta}$ experiment framing	● needs the g_i vector and the T rule (the existing unfinanced autonomous shock is 🔥 as input)	● needs the external-account closure
ALPHA (full employment)	● normalized composition shock runs (the +0.067% η -invariance diagnostic); needs explicit framing and reporting	🔥 the autonomous-demand version is unfinanced; the tax rule is missing	●
BETA (elastic supply)	● closure not implemented	●	●
GAMMA (sticky real wage)	● :fixed implemented; composition run needs a validation pass	● the +19.3 pp extensive-margin sweep exists but rests on the 🔥 unfinanced shock; must be regenerated under F2	●
DELTA (IO endpoint)	● corner not implemented (fixed factor price + Leontief limit)	●	●

Stage 0 remaining items: (i) audit-trail disclosure (brief mention vs. full appendix vs. none; recommended: brief, per the supersession-note culture); (ii) scope locks: skill-class disaggregation (mandatory only for policy-evaluation claims), open-economy depth beyond the section 4.1 mapping (Armington extension optional), B&F replication off the critical path (confirmed by all surviving plans).

4.1 Intermediate results: the cbase2 pipeline (v3 open-economy recalibration) [added v4]

The notebook pipeline `cbase2/` (see `cbase2/documentation.md` for the neutral description and `cbase2/process_comments.md` for the dated observations) has implemented and verified the v3 open-economy recalibration. Three structural findings shaped it. First, the DELTA investigation showed that the fixed-wage rows are indeterminate without marginal leakages: with the import margin alone, the only solution is the degenerate corner $E = 0$, because the import leak has no offsetting injection. Second, the national-accounts identity $S = I + X - M$ forces the saving rate: the household's non-consumption share is the accounting partner of the exogenous investment and export injections, so the v3 calibration derives it from the data ($s = 0.398$; government share $\tau_0 = 0.214$, export share 0.422, investment share 0.162) instead of assuming it. Third, the equilibrium formulation was corrected throughout: ALL N clearing equations are enforced (the N-th market is not Walras-redundant once imports leak), the F1/F3 tax is the fixed nominal baseline budget $T = \sum gG$ (F3 reads exactly as "baseline tax unchanged, programme externally financed"), and F2 keeps the genuinely balanced-budget rule.

Verified results (acceptance test `v2_verify.jl`): the injection continuation runs in 5.4 s with no stalls; the price-explosion branch found by the direct solve is eliminated; the F1/F2/F3 mobile rows solve to residuals of at most $3.7e-7$ with EXACT budget identities ($\sum p c = (1-s)E$); the F3 external balance is recorded ($F = 0.00147$ at the $m = 1$ programme); and the DELTA analytic equivalence is EXACT for both F2 and F3 (rel y error 0.0; $L = 1.1656$ and 1.1767 at the $m = 1$ programme). The updated status matrix, same layout as above:

Labour closure	F1: preference reallocation	F2: tax-financed	F3: external debt
BF (friction, $\eta^* = 0.5$)	● machinery verified with the composition-only stand-in; the explicit $\tilde{\beta}$ tilt experiment pending	✓ implemented and verified (balanced-budget rule; resid $1.1e-7$)	✓ implemented and verified (external balance recorded)
ALPHA (full employment)	● v3 re-run pending ($\eta = 1$)	● v3 re-run pending ($\eta = 1$)	● v3 re-run pending ($\eta = 1$)

Labour closure	F1: preference reallocation	F2: tax-financed	F3: external debt
BETA (elastic supply)	● implemented (η_s , continuation solver; elasticity identification verified at the v2 stage); v3 verification pending	● verification pending	● verification pending
GAMMA (sticky real wage)	● v3 run pending	● the DELTA corner (its Leontief limit) verified exact; the CES-elasticity GAMMA run pending	● v3 run pending
DELTA (IO endpoint, corner)	● F1 cell pending	✓ implemented and verified – EXACT analytic equivalence (rel y error 0.0)	✓ implemented and verified – EXACT analytic equivalence (rel y error 0.0)

Open items carried into Stage 1: the BETA verification (the eta-continuation runtime under v3), the ALPHA and GAMMA v3 re-runs at $\eta = 1$ and the CES elasticities, the explicit F1 tilt, and the interpretive pass on the v3 baseline units (the wage reads 0.54 in the new no-numeraire units while prices stay near 1 – normalization or inconsistency, to be settled in notebook 04).

5 Workplan [section 5 since v4; reworked v3]

5.1 Stage 1: Model completion [reworked v3]

Ordered by the status matrix – the financing core first (it makes every experiment admissible), then the missing labour closures, then the guards:

1. **Financing core:** the shared final-demand and institutional-budget core; the government/investment vector g_i with the tax rule $\sum_i p_i g_i = T$ (F2); the external-account closure with $\sum_i p_i g_i = F$ and import content at fixed sector shares (F3); the explicit preference-reallocation experiment with renormalized weights $\tilde{\beta}_i$ (F1); retire the unfinanced autonomous shock (🔥). – DONE in v3, superseded in scope: the v3 open-economy calibration implements the government block, investment, exports, import margins, the saving rate and all three financings (see section 4, intermediate results). The F1 experiment currently runs with a composition-only stand-in; the explicit $\tilde{\beta}$ tilt is pending.
2. **BETA:** elastic total labour supply on the real wage, $L^s = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$, with the labour-leisure interpretation and the numerical elasticity test $d \log L / d \log (w/P) \approx \eta_s$. – Implemented (η_s , eta-continuation solver, single-point elasticity identification verified at the v2 stage: 0.5/1.0/2.0 recovered); the v3 verification run is pending (eta-continuation runtime).
3. **DELTA corner:** fixed real factor price, unconstrained factor quantity, Leontief limit of the CES core; document the exact equivalence conditions (R2's own point, answered). – DONE: implemented as :fixed + Leontief limit; the analytic equivalence is EXACT for F2 and F3 (rel y error 0.0).
4. **Pre-register** η^* (proposal: 0.5) before any Stage 2 run.
5. **Cobb-Douglas limit guard** (Milestone C): sign-safe real powers at $\varepsilon \rightarrow 1$. – Implemented (analytic branch at $\varepsilon = 1$); the continuous-epsilon assertion is pending its v3 re-run.
6. **Full residual validation** (ROADMAP Phase 4): omitted-equation invariance (rotating the omitted market), household-expenditure exhaustion for every experiment type, homogeneity, multi-start convergence, Tornqvist consistency. Machine-precision residuals are the gate. – Partially done: budget identities exact at every solved equilibrium; ALL N clearing equations enforced in both wage regimes; the omitted-equation rotation and multi-start battery are pending.

5.2 Stage 2: Simulation and sensitivity [reworked v3]

1. **Fill the 5 x 3 matrix:** headline set per cell (Tornqvist real GDP, employment, external balance), sectoral detail into the appendix; compare each cell against its pre-registered expected signature – deviations are findings, not failures.
2. **Sectoral Sobol:** does the reallocation friction (BF) matter for sectoral allocation even where it is aggregate-second-order? (R1.7's aggregate-vs-sectoral demand.)

3. **Robustness:** shock magnitudes; the financing variants are already matrix columns; state the B -collapses-into- T equivalence as a limitation.
4. **Elasticity table** (R1.3): values, ranges, sources, distributions, pre-registered.

Exit criterion (ROADMAP Phase 5): headline rankings qualitatively stable across financing closures and shock sizes; otherwise ranking instability itself is the result.

5.3 Stage 3: Manuscript and reviewer response [reworked v3]

The manuscript is written from the two-wedge design; every referee point is addressed at a named location. The nine-section structure (definitive guide Part III, adapted):

1. **Introduction** – the two wedges; state upfront that the IO–CGE bridge is known (Robinson 2006, Rose 1995); one short Kuhnian paragraph at most (R2.3; R1.1).
2. **Related literature** – the closure taxonomy; Robinson (2006) central; Rose (1995); Dervis–de Melo–Robinson (1982); de Melo & Tarr (1992); McGregor–Swales–Yin (1996); Shoven & Whalley (1984); Mansur & Whalley (1984); Willenbockel (1994) (R2.2; R1.8).
3. **Data and accounting** – the section 4.1 pipeline, import separation, decomposed value added, the documented 5.387% residual, the calibration table (R1.3; R2.4).
4. **Model and closures** – verbal model overview before equations (R1.2); static horizon stated; the two wedges as design dimensions; financing closures explicit (R2.4, R2.5).
5. **Aggregate results** – matrix rows; wage-regime vs elasticity channels (R1.7).
6. **Sectoral results** – sectoral Sobol, bottlenecks (R1.7).
7. **The financing wedge** – bracket results, the external-debt variant, the bounded endogenous-money paragraph.
8. **Robustness and limitations** – financing variants, shock sizes, one-factor aggregation, static horizon, open-economy treatment.
9. **Conclusion** – practical closure guidance; no bridge-discovery claims (R2.1/R2.3 answered by measurement).

Reviewer 1 point map: R1.1 structure/central argument -> sections 1 and 3 (the matrix); R1.2 verbal model description -> section 4; R1.3 calibration summary table -> section 3 and the elasticity table; R1.4 Type-I/II multipliers -> clarify or drop in section 2; R1.5 wage-setting rule -> GAMMA/BETA with explicit floors; R1.6 mobility/unified wage -> BF/ALPHA; R1.7 aggregate vs sectoral -> sections 5–6 with the sectoral Sobol; R1.8 McGregor–Swales–Yin -> literature and the DELTA corner; R1.9 minor copy-edits (p. 5 sentence, p. 16 paragraph, Figure 3 axis).

Reviewer 2 point map: R2.1 labour-supply elasticity as bridging parameter -> BETA and the matrix; R2.2 closure literature -> section 2; R2.3 cleavage framing -> domain-of-applicability reframing, section 1; R2.4 closed-economy miracle -> section 3 pipeline plus the external-debt closure; R2.5 sector-specific immobile labour -> the mobile core (ALPHA/BF); R2.6 eq. (19) sign error -> section 4 fix; R2.7 supply-vs-demand characterization -> removed; R2.8 CES novelty claim -> dropped; R2.9 Cobb–Douglas archetype -> dropped; R2.10 arbitrary uniform allocation -> eliminated by endogenous allocation (BF/ALPHA); R2.11 "fundamentally contested" allocation claim -> dropped or referenced; R2.12 internal mobility contradiction -> resolved by the mobile core; R2.13 McGregor et al. -> section 2 and DELTA; R2.14 skill classes -> optional extension with the limitation stated.

Methodological correction: the corrected-equilibrium story (Result 1 of the definitive guide) is the response letter's opening; the disclosure level remains a Stage 0 decision. Must not appear: all η -dominance claims, the 88.4%/100% figures, GO certification language, price-invariance claims, and "the IO multiplier requires $\eta \rightarrow \infty$ " (it requires sticky wages).

5.4 Stage 4: Response letter and submission [reworked v3]

Reviewer 2: maximally gracious; concede "the bridge exists" fully; present the corrected equilibrium and the measured matrix as the contribution. Reviewer 1: point-by-point via the map above; calibration table; McGregor engagement; aggregate-vs-sectoral quantification. Then the final gates (ROADMAP section 11): clean-checkout reproduction, reconciled accounts, explicit financing, identity tests, sensitivity shares conditional on stated distributions, tag the submission commit.

5.5 Immediate next step [reworked v3]

Stage 0 is resolved except the disclosure decision and the scope locks. Begin Stage 1 with the financing core (g_i , the T rule, the external account, the explicit F1 experiment), then BETA, then the DELTA corner; pre-register η^* before any Stage 2 run.

Update (v4): the financing core and the DELTA corner are done; the remaining Stage 1 items are the BETA verification run, the ALPHA/GAMMA v3 re-runs ($\eta = 1$, CES elasticities), the explicit F1 tilt experiment, and the interpretive pass on the v3 baseline units in notebook 04 – then pre-register η^* and open Stage 2.

The chronological document map, the per-document evaluation of documents 1–10, and the synthesis have been moved to docs/archive/document_review_leftovers.md at Version 3; this working document now contains the foundations, the evaluation matrix, and the workplan only.

6 Revision Log [added v2]

- **Version 1** (September 2026) — Initial chronological assessment of docs/.
- **Version 2** (September 2026) — Fixed a LaTeX build error (backticked paths with underscores inside `\texttt{\textcolor{span}}`); added the chronological roadmap chapter (Stages 0–4) with Decision overview I (labour-market closures) and Decision overview II (financing closures) as Stage 0 tables with recommendations, folding in all still-relevant details from documents 1–8; documents 1–8 moved to docs/archive/ (definitive_guide.md and labor_closures.md remain active); version history block added; roadmap restructured into Foundation I (labour-market closure) and Foundation II (financial closures) with per-option two-column provenance tables (formulation, status, referee point, cost, recommended role, reviewer, old docs, Baqaee–Farhi usage), followed by the workplan; corrected the geometric reallocation row (“used by BF” – the BF 2019 replication ships both the immobile and the mobile variant, verified in the original MATLAB files); added the exogenous labour-endowment shift as a ninth closure table (BF 2022 BLS labour-supply shifts; the original paper’s slack injection); extended the coverage check with firm capacity constraints and adjustment-cost frictions, plus a to-verify nuance on the BF 2019 mobile variant.; added a lineage note under the exogenous-shift table documenting the direct resonance with the original manuscript’s Section 5 (slack narrative borrowed from BF 2022’s social-distancing channel; code mapping of the three slack implementations to the manuscript’s two variants); Foundation I regrouped into the mandatory minimum (reallocation margin, Closure A, sticky wage, Closure D) and the extended option set (Closure B, partial mobility, Closure C, legacy callback, exogenous shift); final selection implemented: 1.1 selected closures BF (explicitly justified vis-a-vis BF 2019’s neoclassical short-run market-friction argument) and ALPHA–DELTA (neoclassical standard, neoclassical labour–leisure, production-agnostic Keynesian, Leontief-inverse emulation) with a pinned suggested narrative; 1.2 notable unselected candidates (partial mobility, complementarity, exogenous shift as scenario device); 1.3 dropped legacy callback; financing deduction routes clarified (household composition vs. government budget vs. lump-sum tax), static-horizon nuance added (debt collapses into the tax variant; foreign financing is the genuinely distinct robustness), recommendation updated to tax principal with preference-reallocation and foreign robustness; header numbering repaired (subsection level under the foundation sections), coverage check moved to Foundation I as its fourth subsection, plausibility parenthetical removed; financing design set per user decision: principal bracket of preference reallocation and tax financing, external-debt (foreign financing) third variant as the static analog of the endogenous-money intuition, imports absorbed mechanically at fixed sector import shares (Armington extension flagged via the existing Omega split); new section “The Evaluation Matrix (5 x 3)” added with pre-registered expected signatures per cell, the limit relations (BF endpoints, BETA to GAMMA, GAMMA plus Leontief = DELTA corner), and the sketch-derived figure “Paradigmatic directions towards the short run” (docs/pictures/paradigmatic_directions.png); Foundation I merged the exogenous-shift table into the Dropped section (one closure, legacy callback = endowment shift); Foundation II restructured into Selected (bracket + external debt) and Dropped (expenditure switching; pure debt subsumed by the tax variant) with a final endogenous-money paragraph; compile-all.zsh repaired (pandoc now called with `–resource-path` so relative image paths resolve against the md directory).
- **Version 3** (September 2026) — Stage 0 answered by the status matrix (5 x 3 with symbols: what exists, what is missing per cell); the second paradigmatic wedge named (financing; loanable-funds vs endogenous-money) with a bounded-strength recommendation; Workplan reworked (Stages 0–4) with inline reviewer-response maps (R1.1–R1.9, R2.1–R2.14) and the two-wedge manuscript structure; chronological review sections moved to docs/archive/document_review_leftovers.md.
- **Version 4** (September 2026) — Intermediate implementation results of the cbase2/ pipeline reported: the Stage 0 chapter promoted to its own section (section 4) with an updated status matrix; the remaining stages form the Workplan section (section 5). New subsection “Intermediate results” records the three structural findings of the v3 open-economy recalibration (the fixed-wage indeterminacy without leakages and the degenerate $E = 0$ corner; the national-accounts identity $S = I + X - M$ forcing the saving rate $s = 0.398$, calibrated from the data together with $\tau_0 = 0.214$, export share 0.422, investment share 0.162; the corrected equilibrium formulation with ALL N clearing equations, the fixed nominal F1/F3 tax and the balanced-budget F2 rule) and the verified results (injection continuation 5.4 s without stalls, price-explosion branch eliminated, F1/F2/F3 mobile rows at $\text{resid} \leq 3.7e-7$ with

exact budget identities, F3 external balance recorded, DELTA analytic equivalence EXACT for F2 and F3). Stage 1 items annotated with their state (financing core and DELTA corner done; BETA implemented with the verification run pending; CD guard implemented; residual validation partially done); immediate next step updated (BETA verification, ALPHA/GAMMA v3 re-runs, explicit F1 tilt, notebook 04 interpretive pass, then pre-registration).